

UCT801: ETHICAL HACKING

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Course Objectives: This course is designed to impart a critical and theoretical and detailed practical knowledge of a range of computer network security technologies as well as network security tools and the services related to Ethical Hacking.

Introduction: Understanding the importance of security, Concept of ethical hacking and essential Terminologies-Threat, Attack, Vulnerabilities, Target of Evaluation, Exploit. Phases involved in hacking.

Footprinting: Introduction to footprinting, Understanding the information gathering methodology of the hackers, Tools used for the reconnaissance phase.

Scanning: Detecting live systems-on the target network, - Discovering services running listening on target systems, Understanding port scanning techniques, Identifying TCP and LIDP services running on the target network, Understanding active and passive fingerprinting.

System-Hacking: Understanding Sniffers, Comprehending Active and Passive Sniffing, ARP Spoofing and Redirection, DNS and IP Sniffing, HTTPS Sniffing.

Session Hijacking: Understanding Session Hijacking, Phases involved in Session Hijacking, Types of Session Hijacking, and Session Hijacking Tools.

Hacking Wireless Networks: Introduction to 802.11, Role of WEP, Cracking WEP Keys, Sniffing Traffic, Wireless DOS attacks, WLAN Scanners, WLAN Sniffers, Hacking Tools, Securing Wireless Networks.

Cryptography: Symmetric and Asymmetric Cryptography, Classical Encryption techniques, Substitution techniques, Block Ciphers Principles, Fiestel Structure, DES, Double and Triple DES, AES, Public Key Cryptography, RSA, Diffie-Hellman Key Exchange, Cryptographic Hash Functions and Digital Signatures.

Laboratory Work:

Lab Exercises including using scanning tools like IPEYE, IPsecScan, SuperScan etc. and Hacking Tools likes Trinoo, TFN2K, Zombic, Zapper etc.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Understand the different phases involved in hacking.
2. Utilize the scanning tools used for the information gathering.
3. Recognize the phases in session hijacking and use the tools for counter-measuring the various sniffing attacks.
4. Analyse different types of attacks on the wireless networks.
5. Describe and apply different types of algorithms for securing the data.

Text Books:

1. Simpson T. M., Backman K., Corley J., Hands-On Ethical Hacking and Network Defense, Delmar Cengage Learning (2011) 2nd edition.
2. Fadia A. and Zacharia M., Network intrusion alert: an ethical hacking guide to intrusion detection, Boston, MA: Thomas Course Technology 3rd edition (2008).

Reference Books:

1. Mathew T., Ethical Hacking, OSB Publication (2003). 2nd edition
2. McClure S., Scambray J. and Kurtz G., Hacking Exposed 7: Network Security Secrets and Solutions, McGrawHill (2012) 7th Edition.